

Modeling Visual Perception & Attention in XR

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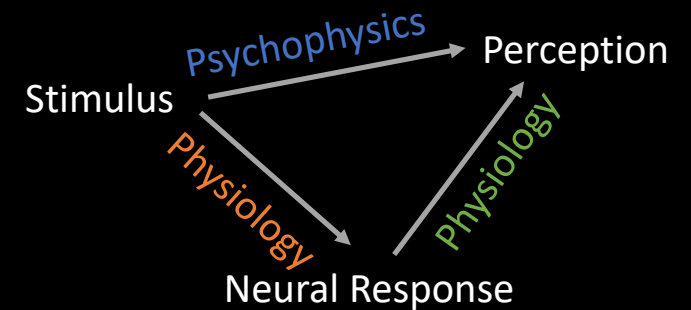
What is perception ?

The acquisition and processing of sensory information in order to see, hear, taste, smell, or feel objects in the world.

An interpretation or impression; an opinion or belief. [Goldstein]

- Approaches

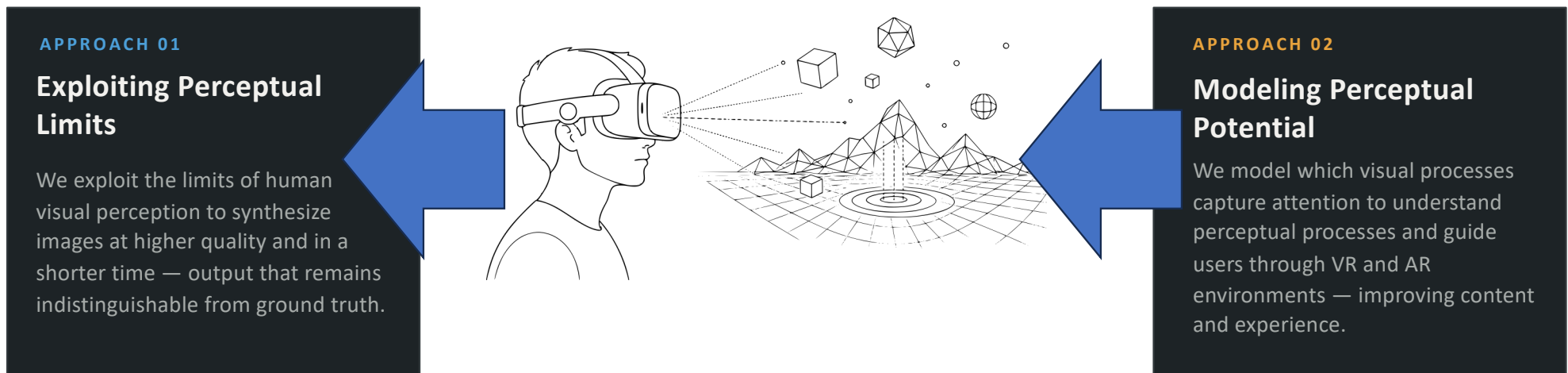
- Psychological – psychophysics
 - Biological – anatomy, physiology
 - Theoretical – descriptive / computational models
- } complementary



Perception-Aware XR

... needs models of perception

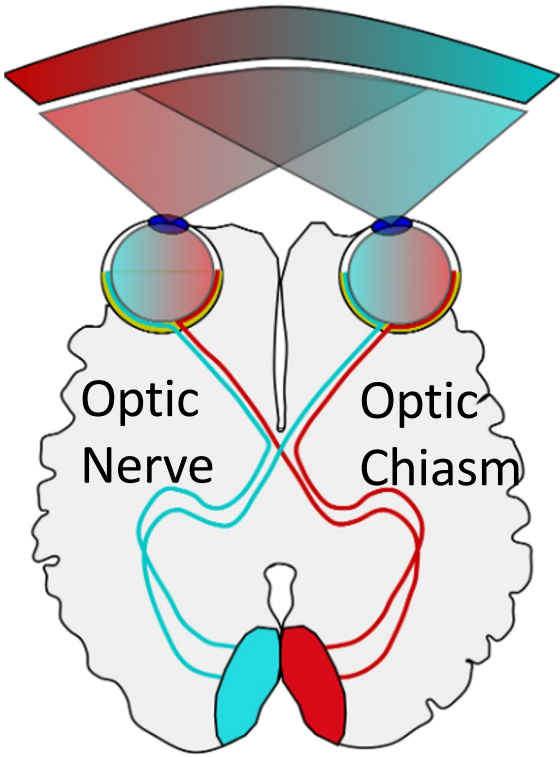
Continuously reason about the limits and the potentials of human vision and use that dual understanding to render smarter and more effectively.



Treat perception as a resource to be actively modeled and exploited on both ends of the pipeline.

Perception-aware XR ... mostly visual

Eyes
Lateral
Geniculate
Nucleus
Primary
Visual
Cortex



	Limitations & Potentials	Low-level Models
FoV	Wide FoV, Stereo, Vergence, ...	IOD/IPD, CVA, ...
Optics Sensor Motor	Optical Pre-Filtering, DoF, Adaptation, Varying Retinal Acuity, Contrast Sensitivity, Ocular Motility (Fixation, SPEM, Saccades), Bleaching, ...	Optical MTFs and Eye Models, Visual Acuity Models, CMF, CSF, Dark/Bright Adaptation Models, Color Appearance Models, TMO, ...
Processing	Crowding, Stereopsis, Cross-modal Integration, ...	Visual Masking, Cortex Transform, Multi-Channel Model, Depth-Cues, ...
Memory & Attention	Object and Spatial Knowledge, Pre-attentive Processing of Patterns, Visual Tunneling and Inattentional Blindness	Singleton Hypothesis, Top-Down-/Bottom-Up Attention Modeling, Attention Steering, Saccade Landing Prediction, ..

Approach 01

Exploiting Perceptual Limitations

Perception-driven Rendering

Visual Computing and Human Perception

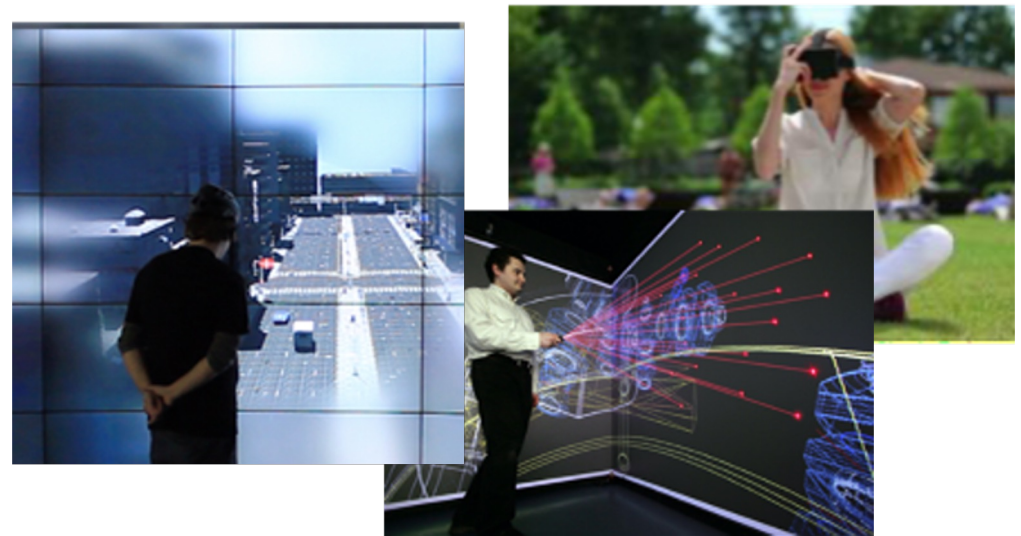
Present a scene in real-time and in a convincing and compelling way computer-generated that cannot be distinguished from the real world

Achieved **realism**, **speed** and **latency** are strongly **limited** by hardware capabilities

We are seeing strong trends towards:

- *Higher pixel densities or larger fields of view, and*
- *More bits per pixel*
- *Increased refresh rates*
- *Increased number of displayed views per pixel*

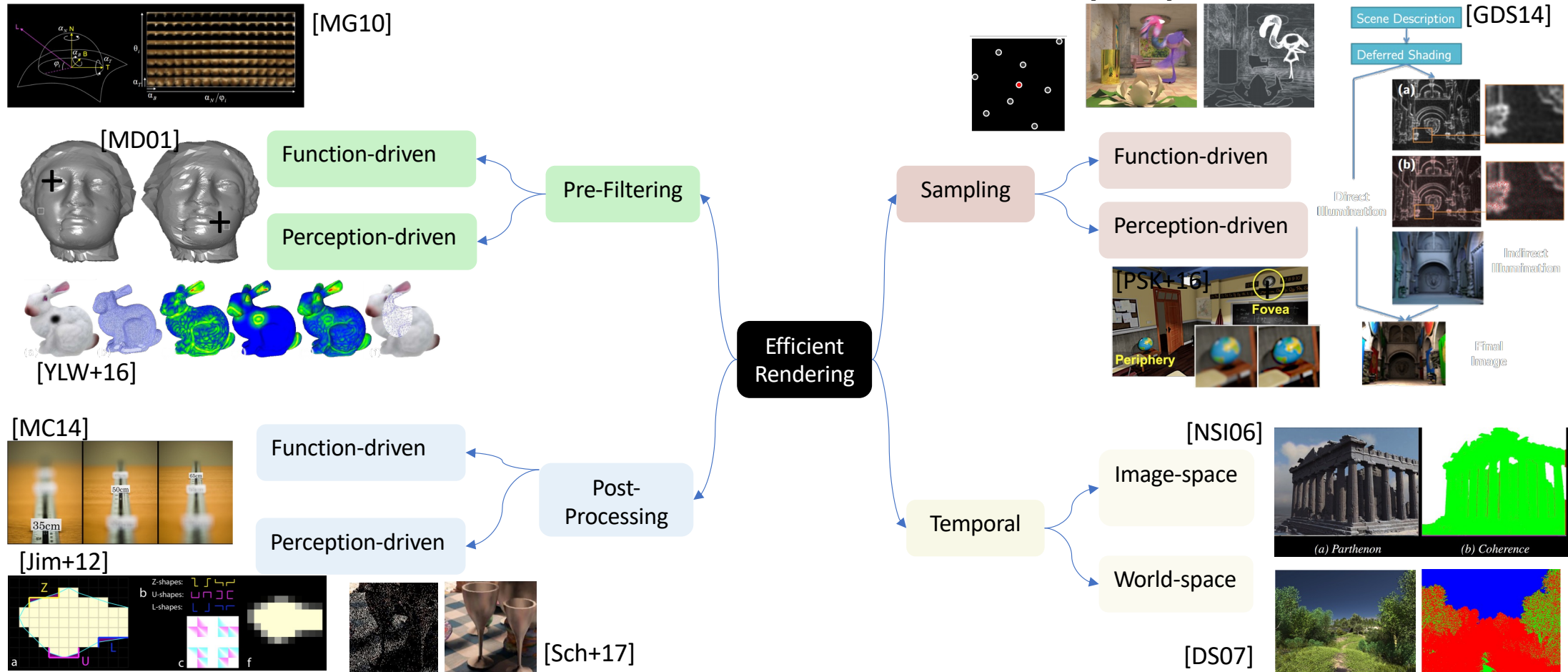
Significant technical challenges for computer graphics!



Exploit the limitations of perception to tackle visual quality and workload

Perception-driven Rendering

Efficient Rendering



Exploiting the Limited Field-of-View

M. Weier, A. Hinkenjann, and P. Slusallek. *A Unified Triangle/Voxel Structure for GPUs and its Applications*. In: *Journal of WSCG*. WSCG 24, (2015), 83–90.

M. Weier, A. Hinkenjann, G. Demme, and P. Slusallek. “Generating and Rendering Large Scale Tiled Plant Populations.” In: *JVRB - Journal of Virtual Reality and Broadcasting* 10.1, 2013.

M. Weier, J. Maiero, T. Roth, et al. *Enhancing Rendering Performance with View-Direction-Based Rendering Techniques for Large, High Resolution Multi-Display Systems*. In: *GI VR/AR*. 2014.

M. Weier, J. Maiero, T. Roth, A. Hinkenjann, and P. Slusallek. *Lazy Details for Large High-Resolution Displays*. SIGGRAPH Asia. 2014. Poster.

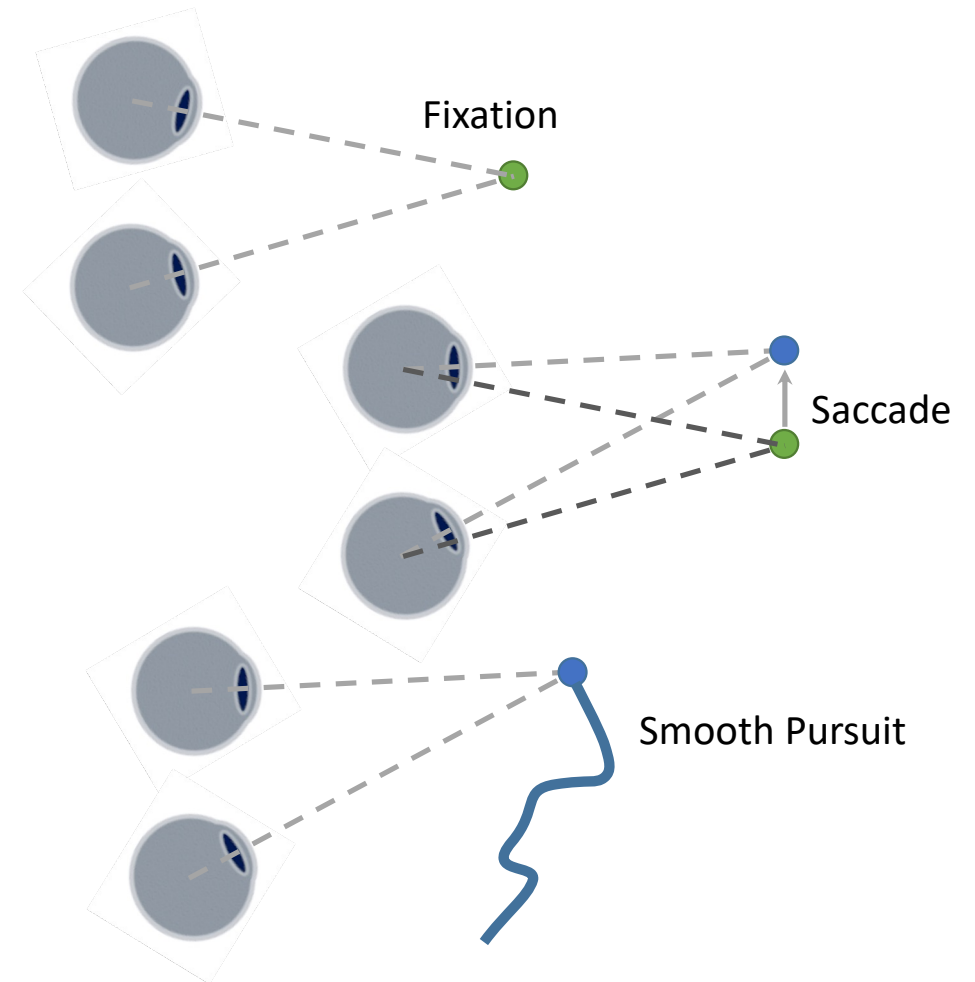


Combine polygonal and voxel information for
a gaze-contingent adaptation of the visual quality

Eye Movements

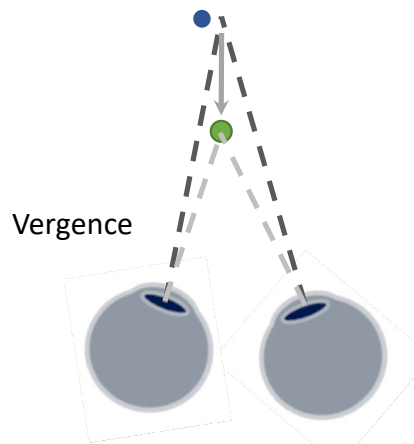
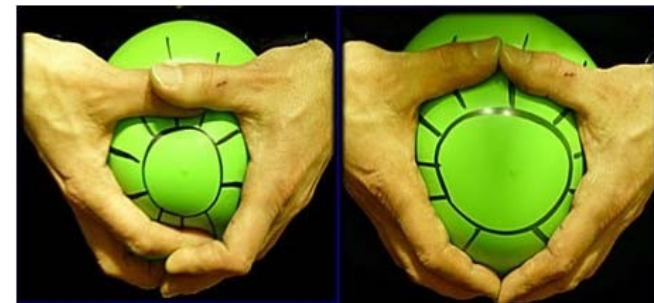
Type	Duration (ms)	Amplitude (1° = 60')	Velocity
Fixation	200-300	-	-
Saccade	30-80	4-20°	30-500°/s
Smooth Pursuit	variable	variable	10-30°/s
Vergence	300-600	-	-
Microsaccade	10-30	10-40'	15-50°/s
Tremor	10	<1'	20'/s
Drift	200-1000	1-60'	6-25'/s

typical
eye
tracker



Eye Movements

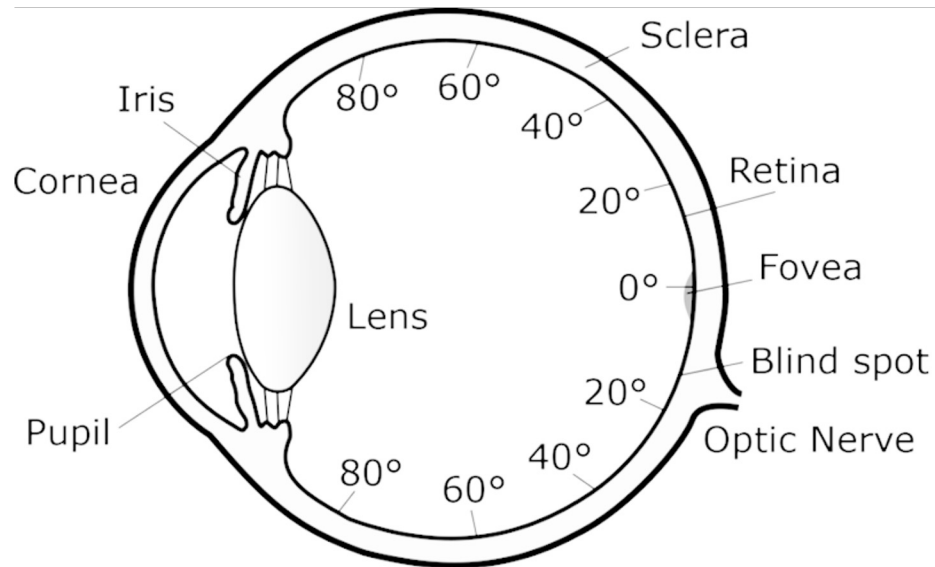
- Accomodation (Duration: 200 ms)
- Convergence and divergence



Quelle: i-fidelity.net

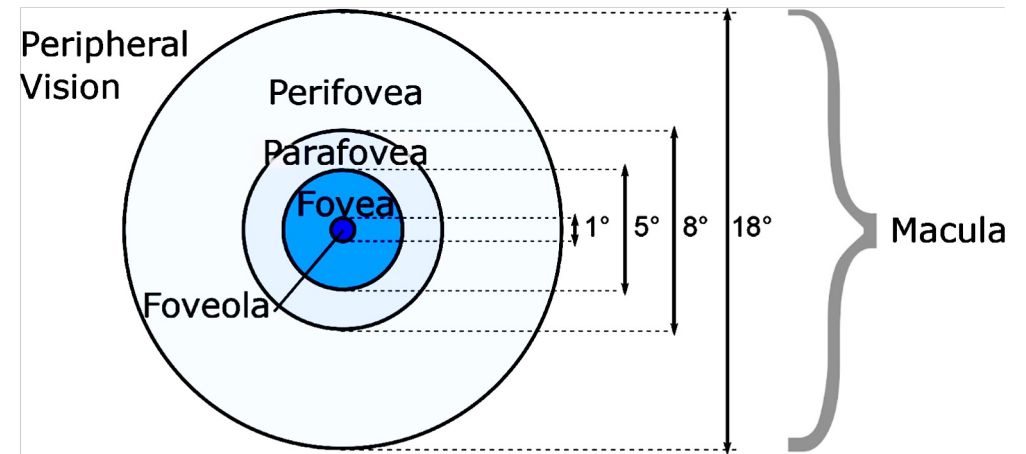
Accomodation

Optics : Monocular Vision

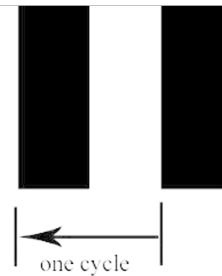


0.55 | 3.05 | 4.0 | 4.32 | 12.0 mm

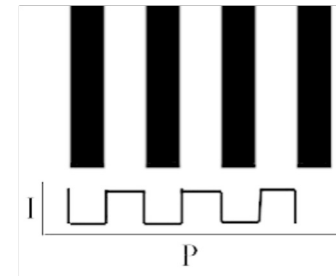
cut-off frequency (given by lens properties) of grating pattern = 87 cycles per degree (of visual angle) → finer details cannot be resolved



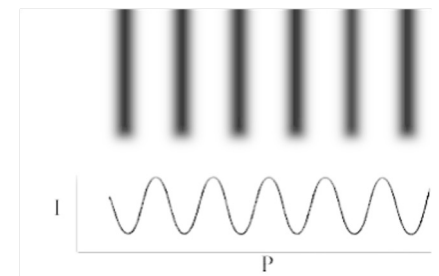
Grating



Square-wave

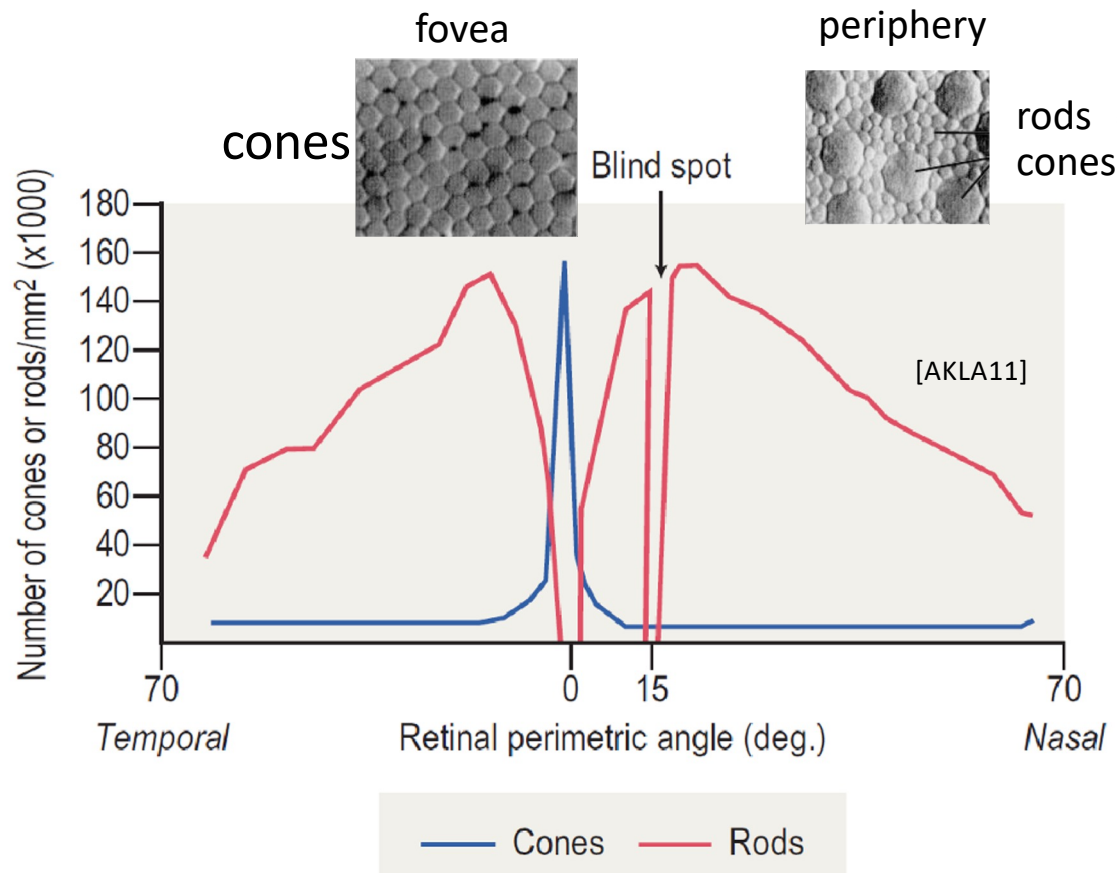


Sine-wave



Visual Acuity

As a function of number of rods and cones



Minimum visible acuity : 0.0084'

detection of a **feature**

Minimum resolvable acuity : 1.02'

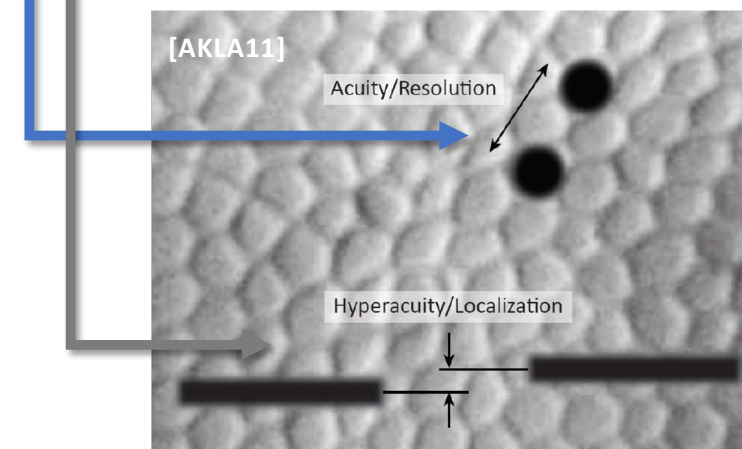
resolution of **two features**

Minimum recognizable acuity : 1.02'

identification of a feature

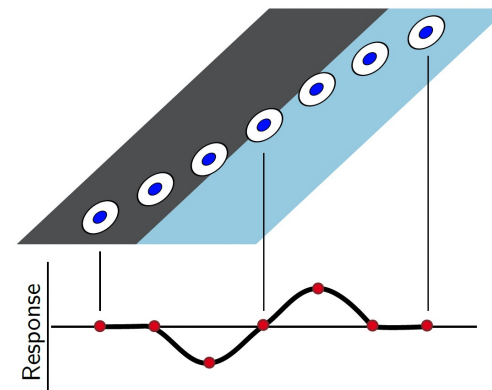
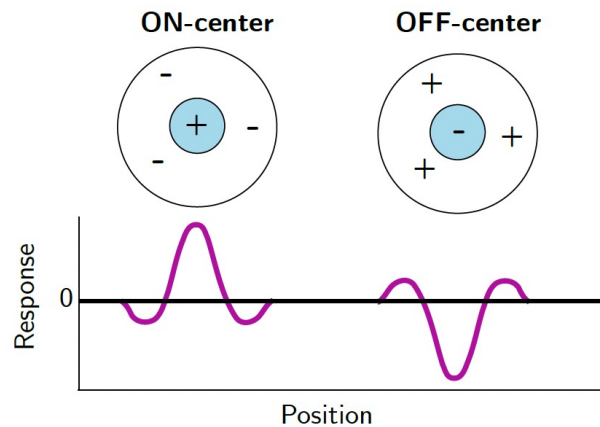
Minimum discriminable acuity : 0.0144'

discrimination of a **change** in a feature



Visual Acuity

As a receptive fields



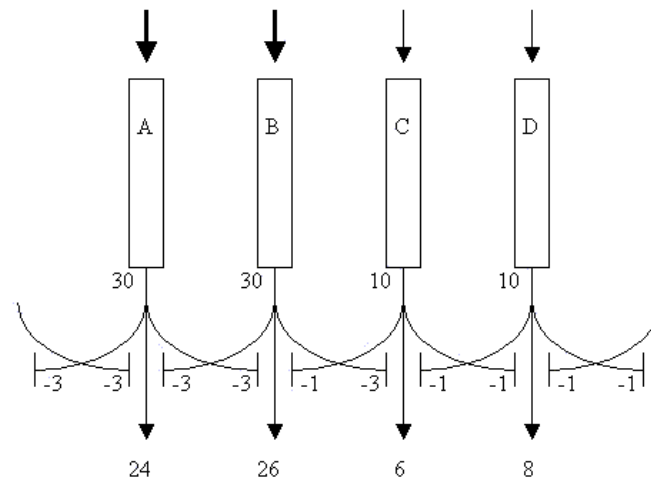
A moving high-contrast edge produces

=> a continuous sequence of on/off responses across the retina

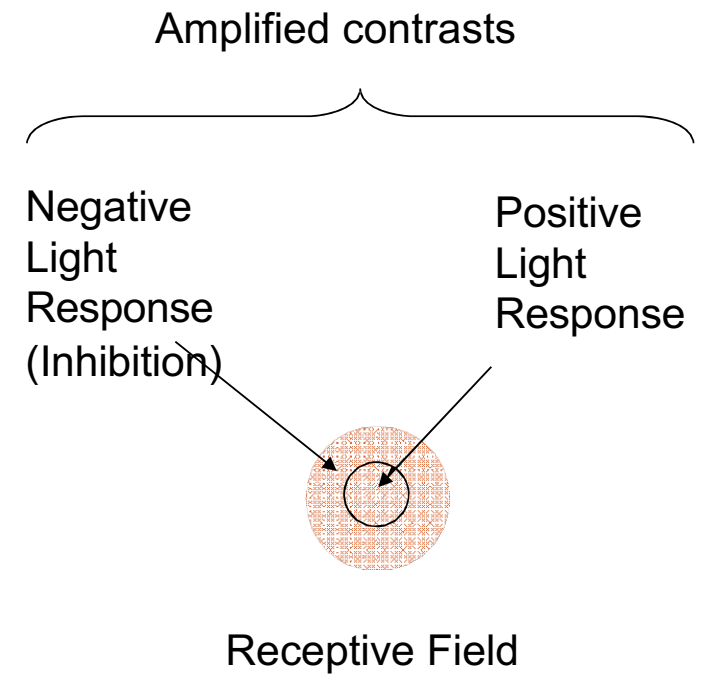
=> a strong, coherent change in luminance across many neighboring receptive fields

Contrast Sensitivity

Mach Bands, Hermann Grid and Receptive Fields

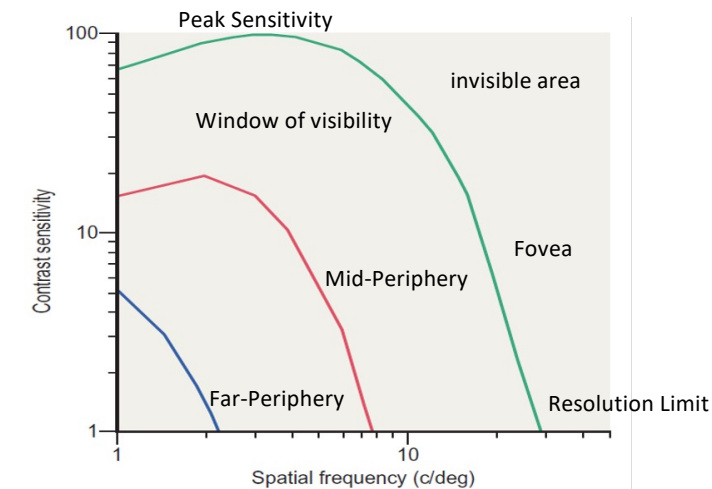
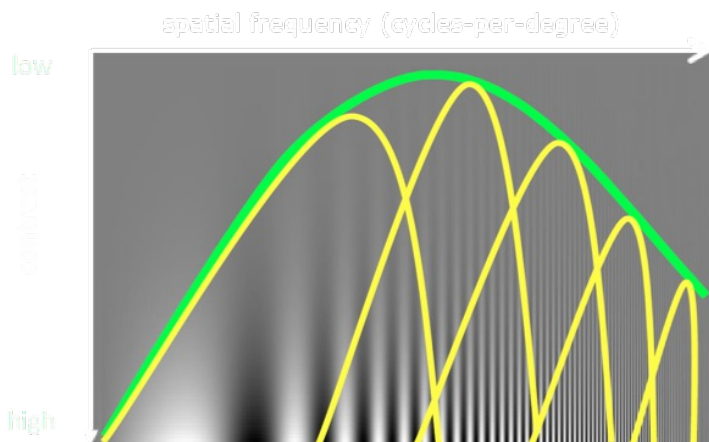


Receptive Field with negative inhibition
(Lateral Inhibition)



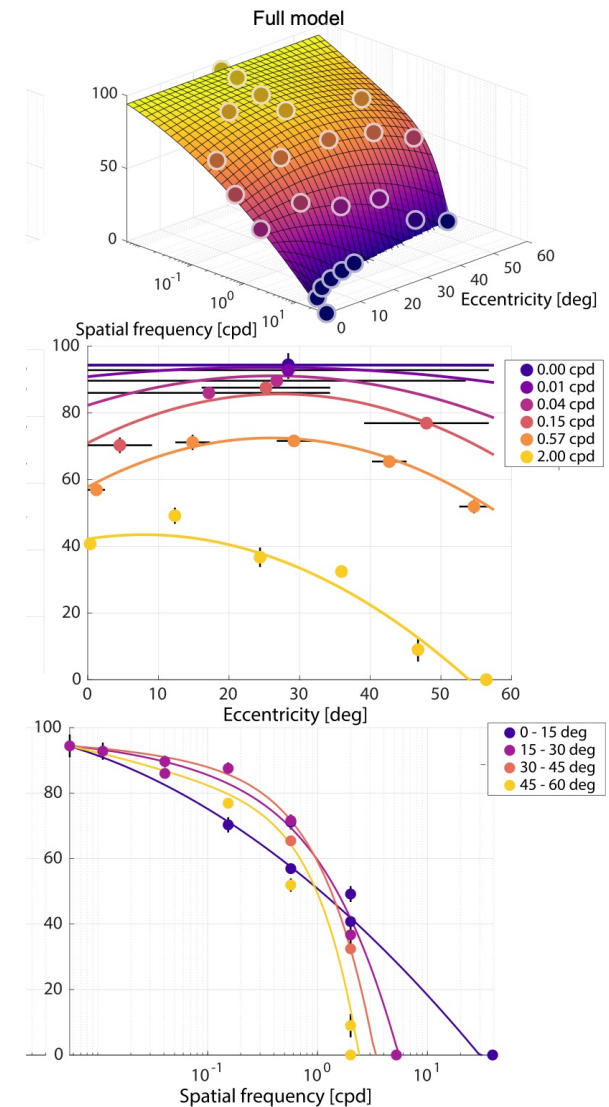
Visual Acuity

As sensitivity to contrasts



CSF is dependent on e.g. **eccentricity, adaptation, orientation, retinal velocity, chromatic and achromatic stimuli**

=> **Visual acuity gets poor, but sensitivity to contrast at low spatial frequencies or during rapid temporal changes is maintained.**



Krajancich, B., Kellnhofer, P., and Wetzstein, G. 2021. A perceptual model for eccentricity-dependent spatio-temporal flicker fusion and its applications to foveated graphics. ACM Transactions on Graphics (TOG) 40, 4, Article 47

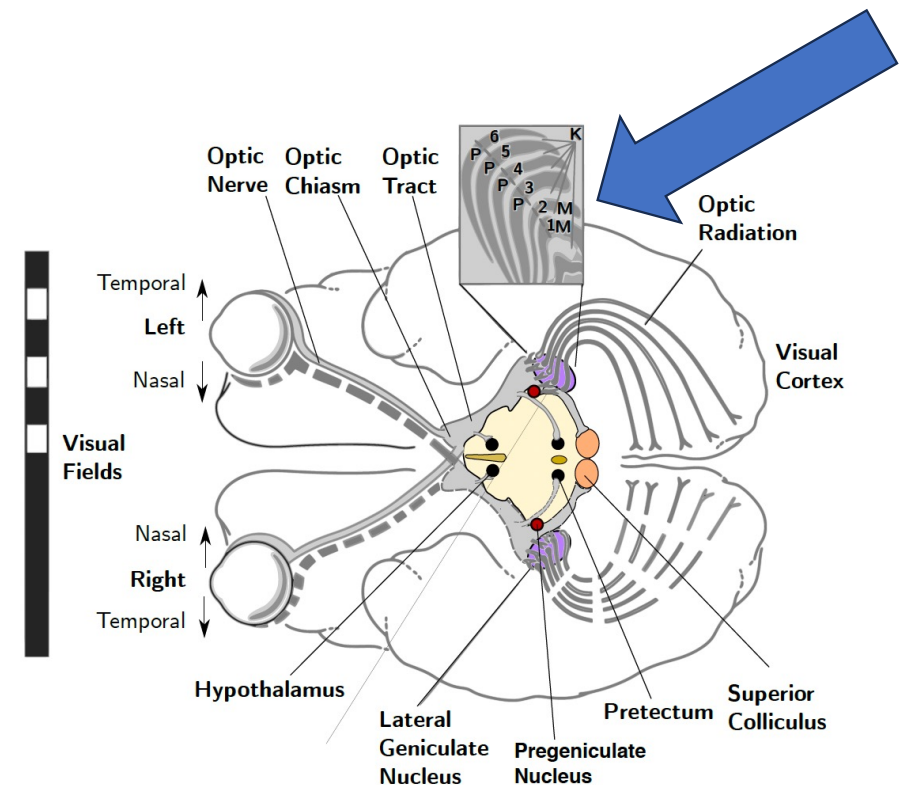
Visual Acuity

Further along the line - Magnocellular Pathways

Peripheral vision feeds strongly into the magnocellular pathway (LGN), which is especially sensitive to:

- high luminance contrast
- low to medium spatial frequencies (broad edges rather than fine details)
- temporal changes and motion

A moving contrast edge is essentially “ideal stimulus” for M-pathway neurons, so it is perceived very saliently even far from the fovea.



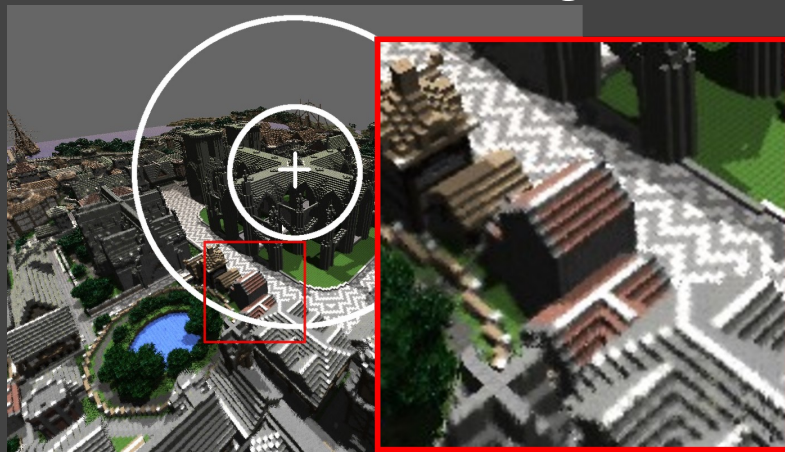
Foveated Rendering

M. Weier, T. Roth, E. Kruijff, A. Hinkenjann, A. Pérard-Gayot, P. Slusallek, and Y. Li. *Foveated Real-Time Ray Tracing for Head-Mounted Displays*. In: *Computer Graphics Forum (Pacific Graphics '16)*. 2016.

T. Roth, **M. Weier**, A. Hinkenjann, Y. Li, and P. Slusallek. "A Quality-Centered Analysis of Eye Tracking Data in Foveated Rendering." In: *Journal of Eye Movement Research (JEMR)* 10.5 (2017).

A. Pérard-Gayot, **M. Weier**, R. Membarth, P. Slusallek, R. Leiða, and S. Hack. "RaTrace: Simple and Efficient Abstractions for BVH Ray Traversal Algorithms." In: *Proceedings of the Conference on Generative Programming: Concepts & Experiences (GPCE)*.

Foveated Rendering



Conventional Rendering



Adapt sampling in a real-time ray tracer to the retinal limitations,
i.e. foveated rendering

Foveated Real-time Ray Tracing

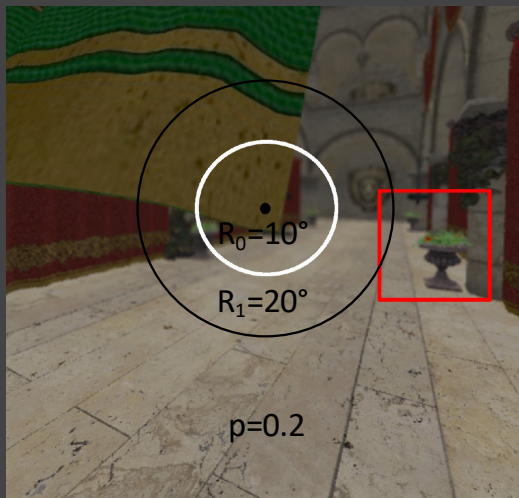
M. Weier, T. Roth, E. Kruijff, A. Hinkenjann, A. Pérard-Gayot, P. Slusallek, and Y. Li. *Foveated Real-Time Ray Tracing for Head-Mounted Displays*. In: *Computer Graphics Forum (Pacific Graphics '16)*. 2016.



Exploiting the Limitation of the Optics

M. Weier, T. Roth, A. Hinkenjann, and P. Slusallek. "Foveated Depth-of-Field Filtering in Head-Mounted Displays." In: *ACM Transactions on Applied Perception (TAP)*. Aug. 2018.

M. Weier, T. Roth, A. Hinkenjann, and P. Slusallek. "Predicting the Gaze Depth in Head-mounted Displays using Multiple Feature Regression." In *ACM ETRA*. June 2018.



Without DoF



Full



Foveated

With DoF



Full



Foveated

DoF is applied in a post-processing step to conceal visual artifacts by removing high-frequency signals from the visual periphery using the inherent blur of the DoF

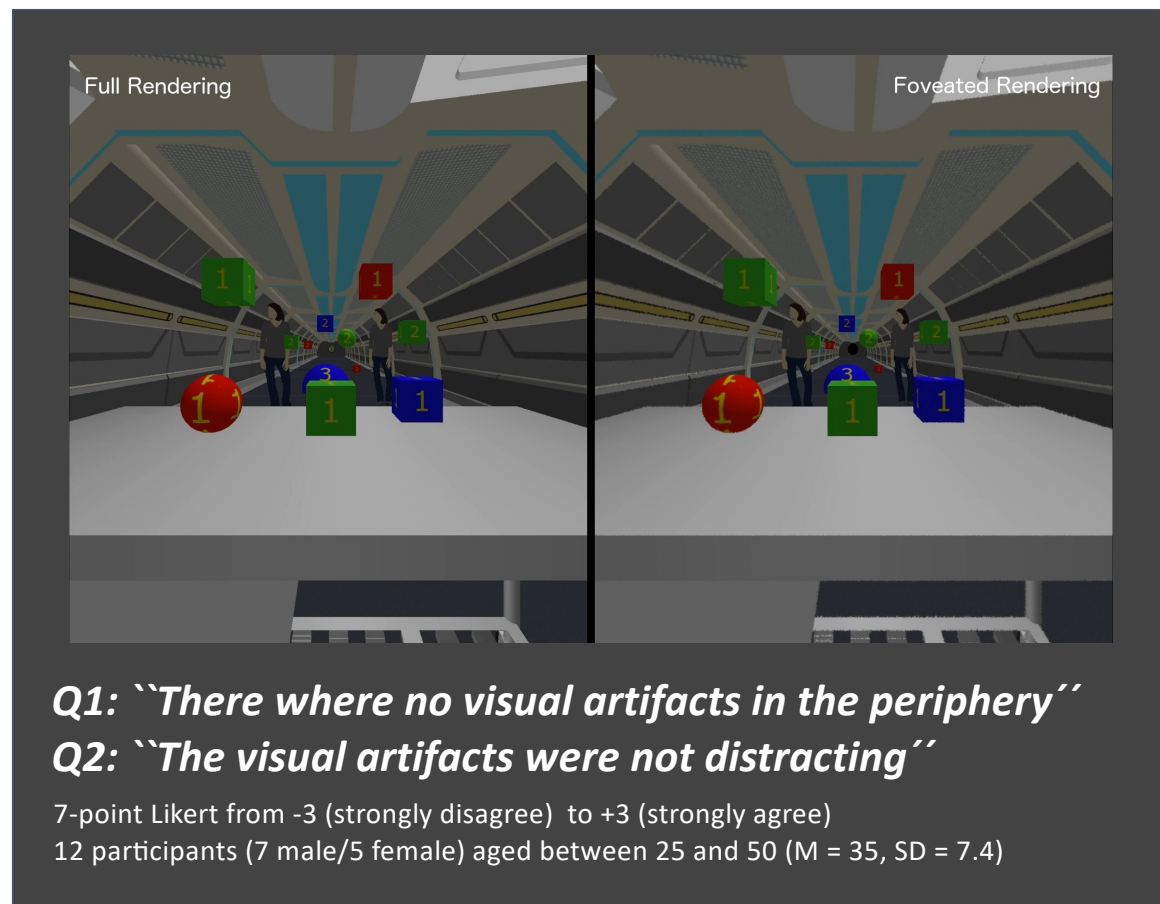
Benchmarks and User Study

Benchmarks

	Rendering	DoF	Total	#Samples
Ours	5.547ms	2.1ms	7.64ms	558,945
Full	6.61ms	2.07ms	8.68ms	1,843,200

Reduces sampling rate by a factor of 3-4

Intel Core i7-3820 @ 3.6GHz, 16GiB RAM,
2 x NVIDIA GeForce GTX 1080 Ti



Sensor: Color Sensation

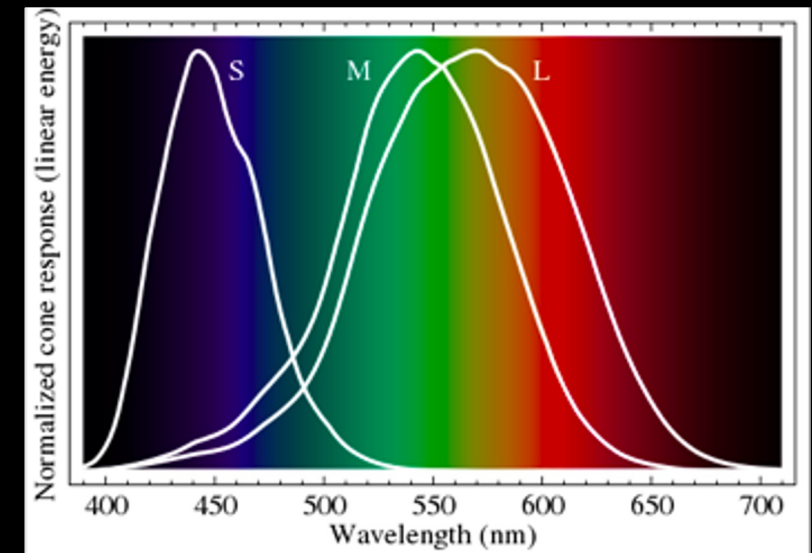
Three different cone types :

- **L(ong), M(edium), S(hort) cones** have different cone response functions
→ spectral sensitivity $l(\lambda)$, $m(\lambda)$, and $s(\lambda)$

$$(L, M, S) = \int_{390}^{750} \underbrace{l(\lambda)}_{\text{spectral sensitivity}}, \underbrace{m(\lambda)}_{\text{spectral sensitivity}}, \underbrace{s(\lambda)}_{\text{spectral sensitivity}} \underbrace{I(\lambda)}_{\text{illumination}} \underbrace{R(\lambda)}_{\text{scene reflectance}} d\lambda$$

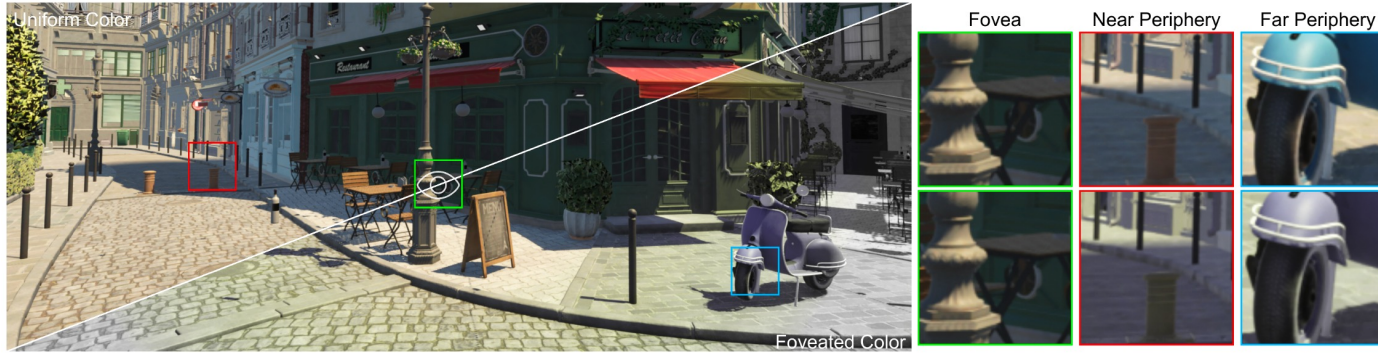
stimulus response

- forms basis for **color matching** and creation of **color spaces**
→ **different** spectral light compositions produce the **same perceived color**
- Cells not evenly distributed on retina



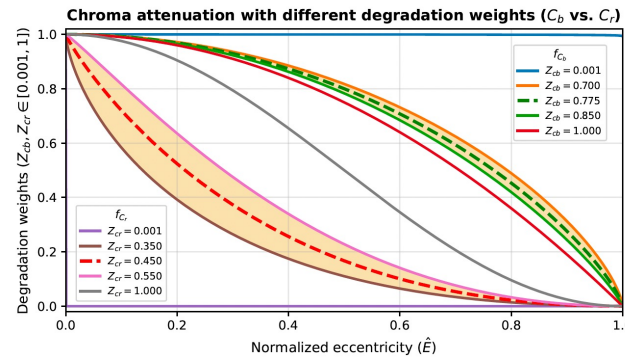
Courtesy of BenRG

Gaze-contingent color adaptation

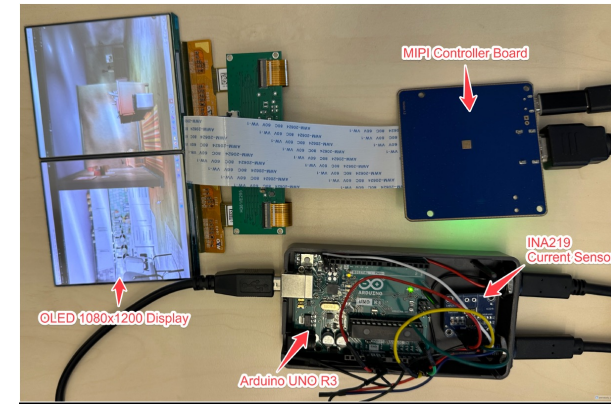
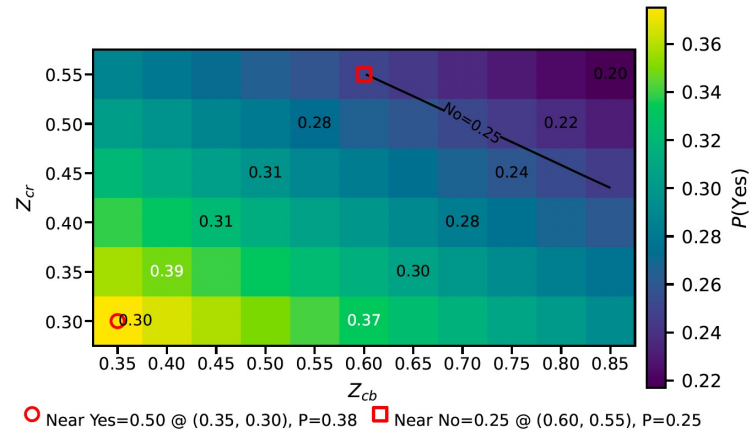


Perceptually calibrated degradation weights for the YC_bC_r color space

$$f_{C_b}(\hat{E}; Z_{cb}) = \left(1 - \hat{E}^2\right)^{Z_{cb}}, \quad f_{C_r}(\hat{E}; Z_{cr}) = \frac{1}{2} \left[1 + \cos\left(\pi \hat{E}^{Z_{cr}}\right)\right], \quad 0 \leq \hat{E} \leq 1.$$



Gaze-contingent color adaptation



Scene	Uniform H.264 (MB)	Foveated mean H.264 (MB)	Δ H.264 (%)	Uniform HEVC (MB)	Foveated mean HEVC (MB)	Δ HEVC (%)
Emerald Square	37.451	34.028	-9.14	10.195	9.171	-10.04
Modular Victorian City	46.662	45.168	-3.20	9.773	9.507	-2.72
Stylized House Interior	30.192	29.337	-2.83	7.026	6.843	-2.60
Amazon Bistro	27.158	26.677	-1.77	8.290	8.084	-2.48
Sun Temple	10.697	10.673	-0.22	3.858	3.787	-1.83
Gwangju Theater	237.168	234.241	-1.23	60.163	60.145	-0.03
Mean over scenes	64.888	63.354	-3.07	16.551	16.256	-3.28

Mohanto, B., Weier, M., Groth, C., Kudnick, J., Marquardt, A., Bruckner, S., & Staadt, O. (2026). Perceptually calibrated foveated chroma attenuation for gaze-contingent VR rendering and displays. To appear in Proceedings of the ACM Symposium on Virtual Reality Software and Technology (VRST '26).

Lesson learned

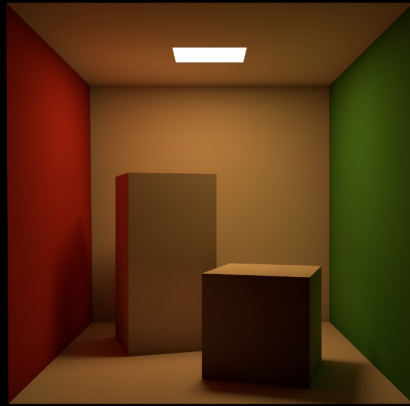
Peripheral vision is built to be a motion-and-contrast alarm system but is highly selective and perceptual resources are shared among senses

The perception of a virtual world cannot be explained by a single physiological model

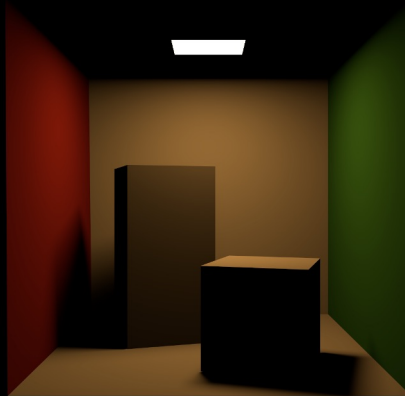
High-level Perceptual Models

Full Reference Metrics

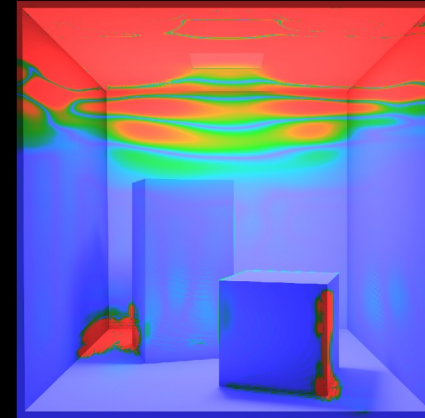
Compare images against a reference to compute a perceptual difference (map)



Reference BDPT



Test Direct Illum.



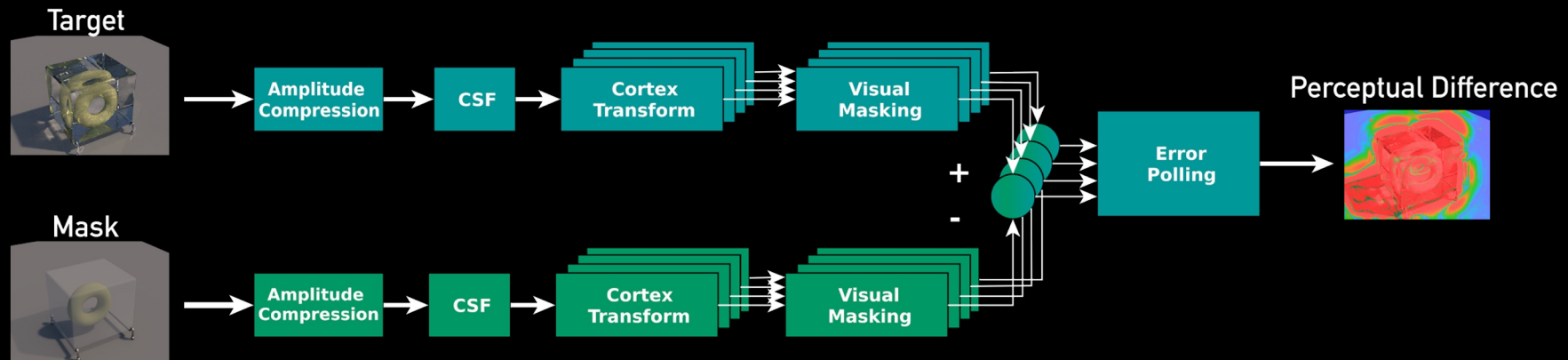
Perceptual Difference Map
(VDP)

- Challenging to use to guide rendering
 - Compare subsequent rendering steps

Full Reference Metrics

VDP has come a long way...

- Daly's Visible Differences Predictor (VDP) [Dal93]



- Jin et al. [JFN98] and Tolhurst et al. [TRP*05] included eye's color sensitivity
- Mantiuk et al. [MDMS05,MKRH11] included HDR (HDR-VDP/HDR-VDP2)
- Narwaria et al. [NPDSLCP14] employ a database of HDR images and their subjective ratings to improve quality
- Swafford et al. [SIGK*16] add quality degradation for peripheral vision

[Dal93] **Daly**: *Digital Images and Human Vision*. MIT Press, 1993, ch. The Visible Differences Predictor: An Algorithm for the Assessment of Image Fidelity, pp. 179–206

[JFN98] **Jin, Feng, Newell**: *The development of a color visual difference model (CVDm)*. IS&T's PICS Conference '98, pp. 154–158

[TRP*05] **Tolhurst et al.**: *A Multiresolution Color Model for Visual Difference Prediction*, APGV '05, ACM, pp. 135–138

[MDMS05] **Mantiuk et al.**: *Predicting Visible Differences in High Dynamic Range Images: Model and its Calibration*. In *Electronic Imaging 2005*, pp. 204–214

[MKRH11] **Mantiuk et al.**: *HDR-VDP-2: a calibrated visual metric for visibility and quality predictions in all luminance conditions*. In *ACM TOG* (2011), vol. 30

[NPDSLCP14] **Narwaria et al.**: *On Improving the Pooling in HDR-VDP-2 towards Better HDR Perceptual Quality Assessment*. In *Human Vision and Electronic Imaging '14*, pp. 1–6

[SIGK*16] **Swafford et al.**: *User, Metric, and Computational Evaluation of Foveated Rendering Methods*. *ACM SAP '16*, pp. 7–14

Full Reference Metrics

VDP has come a long way...

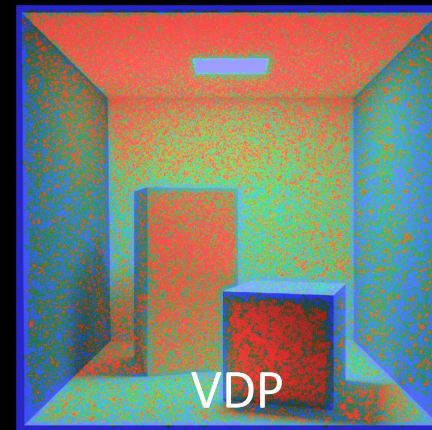
- + Good for detecting rendering artifacts, such as MC noise
- + Good repeatability of results
- Absolute measure of the image quality is difficult to obtain
- Limited when determining the magnitude of clearly visible (supra-threshold) artifacts
- Too computationally demanding for real-time use



Reference 256spp



Test 4spp



VDP

No-Reference Metrics

In computer graphics images are synthesized. There is no reference!

No-Reference Metrics

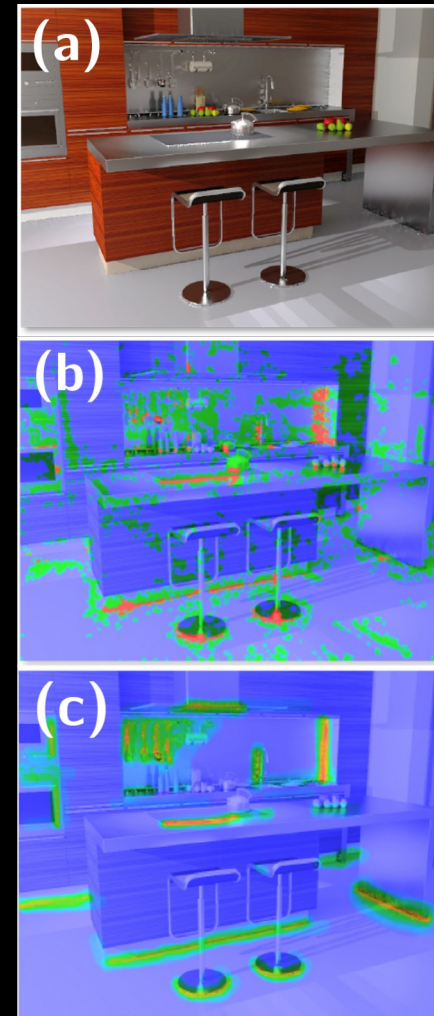
- Blindly estimate the quality of the image or an animation sequence
- Enable to directly judge the quality of single images

Early methods

Chandler and Hemami [CH07] quantify the visual fidelity of images based on VSNR

1. Contrast thresholds for detection of distortion
2. Wavelet-based models of visual masking/summation
3. Low-level vision model to accommodate viewing conditions, contrasts, etc.

**HUMAN-
CENTRED**



Perception-based Fidelity Metrics

Current trends

Čadik et al. [ČHM*12] compare image quality metrics in detecting rendering artifacts and provide a dataset



Ref.



Artf



Study

VDP

• We still perform suboptimal!

Machine learning:

- Already successful in predicting a single mean opinion score (MOS) [ZK15, ZWF16, APY16].
- High learning curve but cheap to evaluate

[APY16] Amirshahi et al.: *Image quality assessment by comparing cnn features between images*. Journal of Imaging Science and Technology 60, 6 (2016)

[ČHM*12] Čadik et al.: *New Measurements Reveal Weaknesses of Image Quality Metrics in Evaluating Graphics Artifacts*. ACM TOG SIGGRAPH Asia 2012, 31, 6 (2012), pp. 147–157

[ZK15] Zagoruyko, Komodakis: *Learning to compare image patches via convolutional neural networks*. In Proc. of the IEEE Conf. on Comp. Vis. and Patt. Rec. (2015), pp. 4353–4361

[ZWF16] Zuo, Wang, Fu: *Screen content image quality assessment via convolutional neural network*. IEEE ICIP '16, pp. 2082–2086

Ideas for Open Research Topics

- Gaze-contingent acuity, color (e.g. dkl) or attention models for
 - Rendering speed/quality enhancements (e.g. with Motion Metamers)
 - Compression
 - Power Reduction
- Gaze-contingent rendering for Gaussian Splatting, Holography, Light Fields, ...
- Foveated denoising for real-time path tracing
- Foveated no-reference image quality metrics
- Combine gaze-guidance and scan path prediction to steer rendering quality
- Perceptual metrics in generative AI systems
- ...

Tariq, T., & Didyk, P. (2024). Towards Motion Metamers for Foveated Rendering. ACM Transactions on Graphics, 43(4), 1–10. <https://doi.org/10.1145/3658141>

Approach 02

Modeling Perceptual Potential

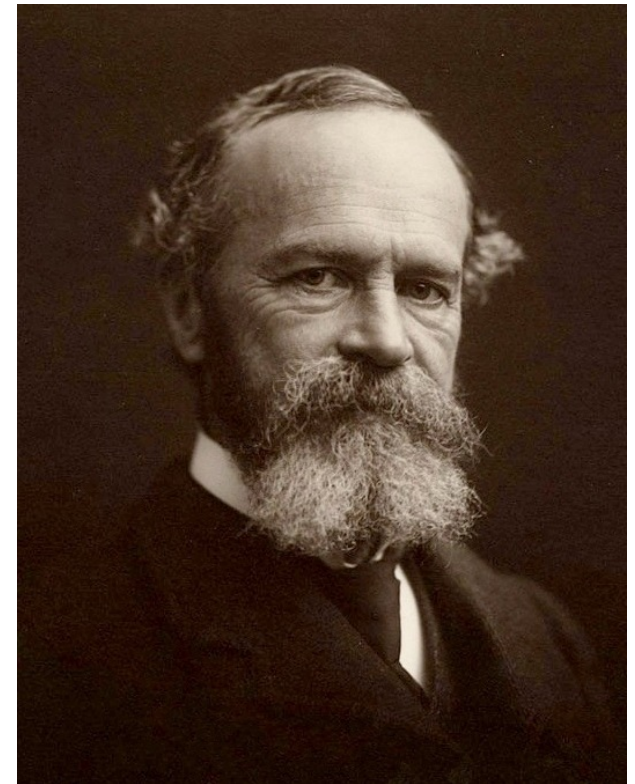
WHODUNNIT?



What is attention?

“Everyone knows what attention is. It is the taking possession by the mind, in clear and vivid form, of one out of what seem several simultaneously possible objects or trains of thought...

... It implies withdrawal from some things in order to deal effectively with others”



**William
James**

W. James (1890). *The principles of psychology*.

What is attention?

The cognitive process of **selectively concentrating on one aspect** of the environment while ignoring other things.

What is attention?

The cognitive process of **selectively concentrating on one aspect** of the environment while ignoring other things.



Anderson (2004). *Cognitive psychology and its implications*.

How do we direct attention?

Overt visual attention: Involving eye movements

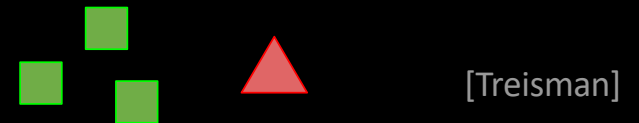
Covert attention: Without eye movements (not easily observable)



Overt **shifts of attention** are mainly associated with the execution of **saccadic eye movements**

Memory : Attention

- brain provides limited processing capacity
- **attentional spotlight** adjusts **processing power** [Crick84]
- **pre-attentive** processing
bottom/up, stimulus-driven
- **attentive** processing
top-down / context-driven, goal-driven
- features attracting attention : **motion**, **flickering**, **color**, **(known) objects**, **faces**, etc.
- **Crowding** : object visibility in periphery changes based on spacing distances
- further concepts : **Selective Attention**, **Change Blindness**

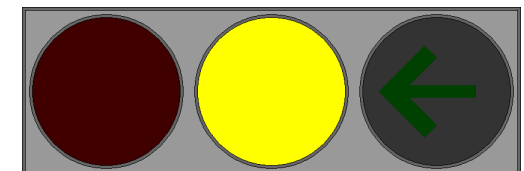
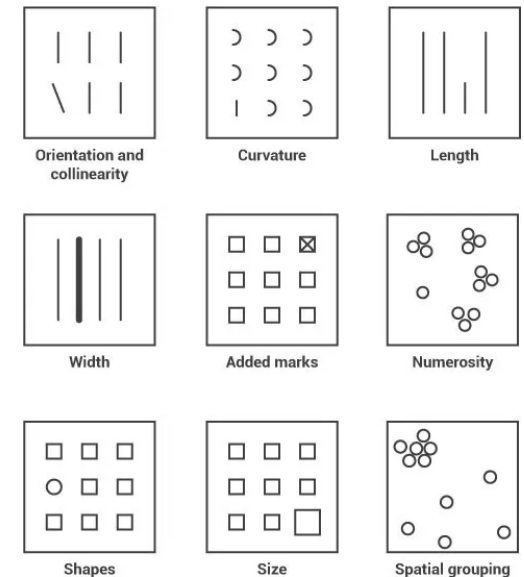
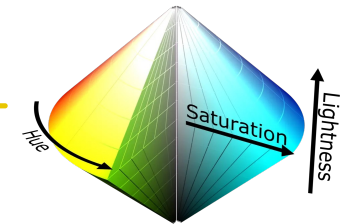


Pre-attentive Perception

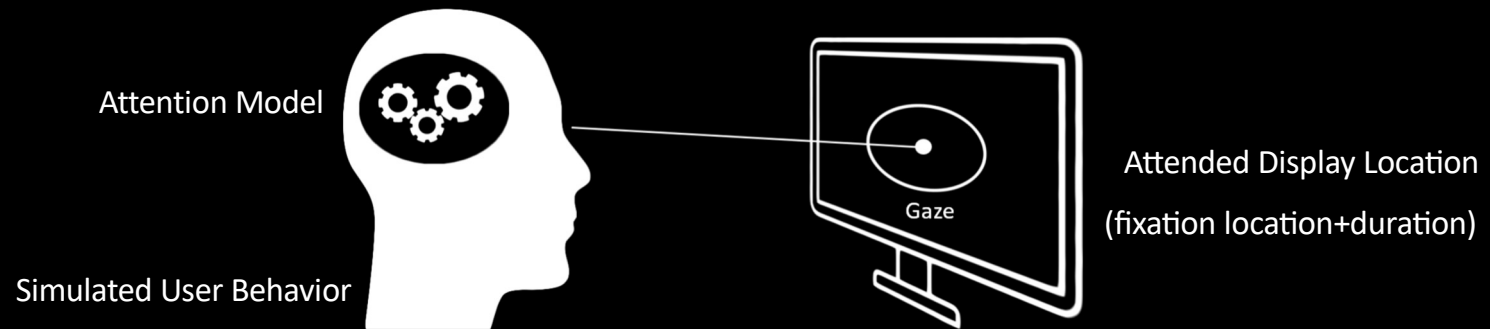
“The greatest value of a picture is when it forces us to notice what we never expected to see.” – John Tukey

Preattentive Features are processed in parallel and do not require our active attention

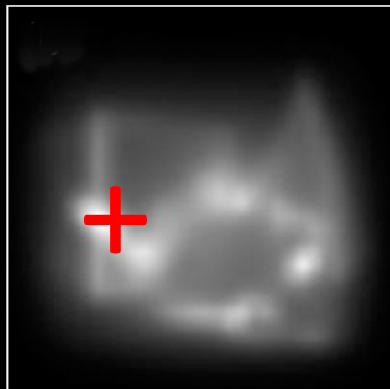
- Color
- Shape
- Movement
- Spatial Positioning



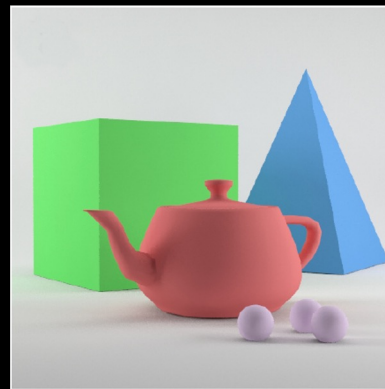
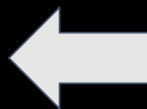
Modeling Attention



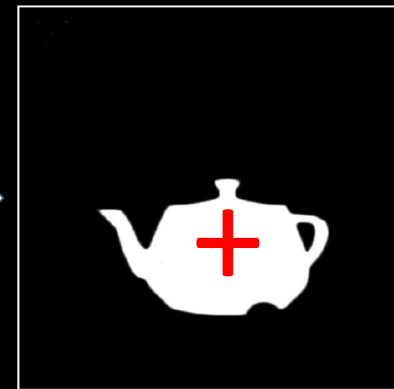
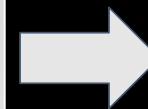
Bottom-up Model



Saliency Map



Input Image



Task Map

Top-Down Model

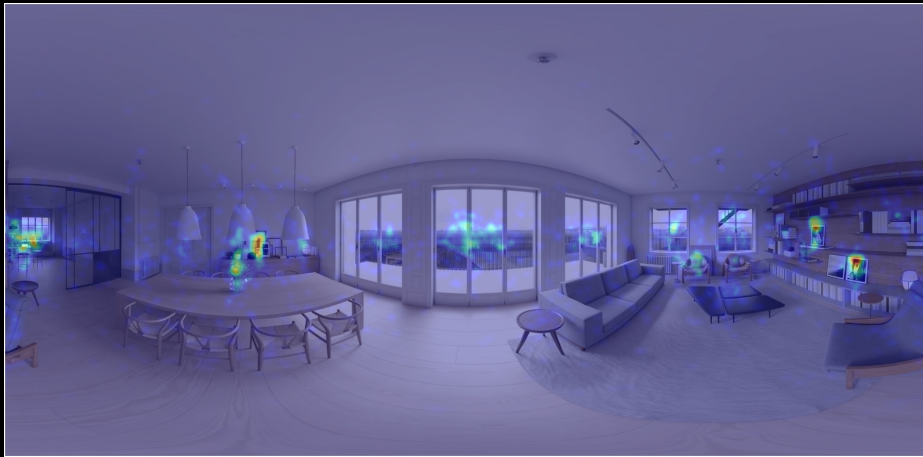
**HUMAN-
CENTRED
PERCEPTION.**

[BI13] Borji, Itti: *State-of-the-art in visual attention modeling*. IEEE Transactions on Pattern Analysis and Machine Intelligence (2013)

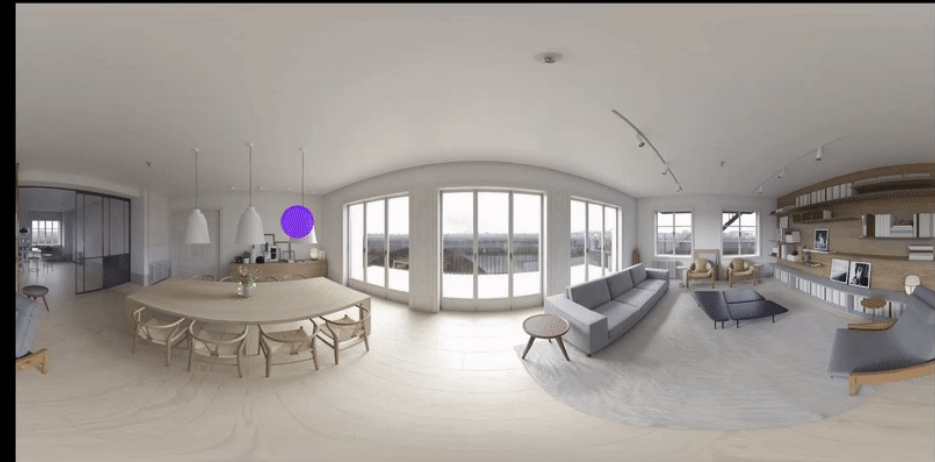
Modeling visual attention

Computational representations of attention

Saliency Maps



Scan Paths



Computational models of attention Taxonomy*

Computational models of attention Taxonomy*

Biologically-inspired models

Try to **emulate human neural mechanisms** to predict attention, replicating how the brain processes visual information.

Computational models of attention Taxonomy*

Biologically-inspired models

Try to **emulate human neural mechanisms** to predict attention, replicating how the brain processes visual information.

Probabilistic models

Originate from **information theory**, where attention is predicted based on self-information metrics like rareness, surprise, and contrast.

Computational models of attention

Taxonomy*

Biologically-inspired models

Try to **emulate human neural mechanisms** to predict attention, replicating how the brain processes visual information.

Probabilistic models

Originate from **information theory**, where attention is predicted based on self-information metrics like rareness, surprise, and contrast.

Data-driven models (learning based)

Leverage large datasets and machine learning techniques to learn patterns of attention **directly from empirical data**.

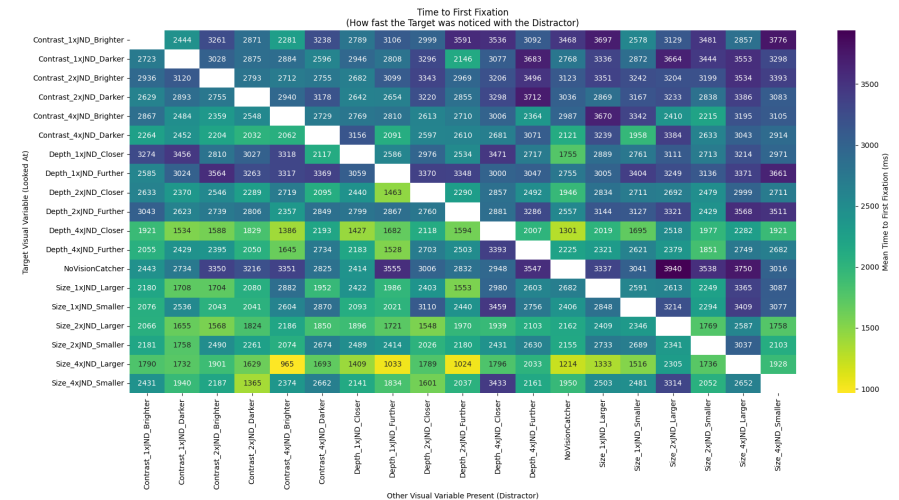
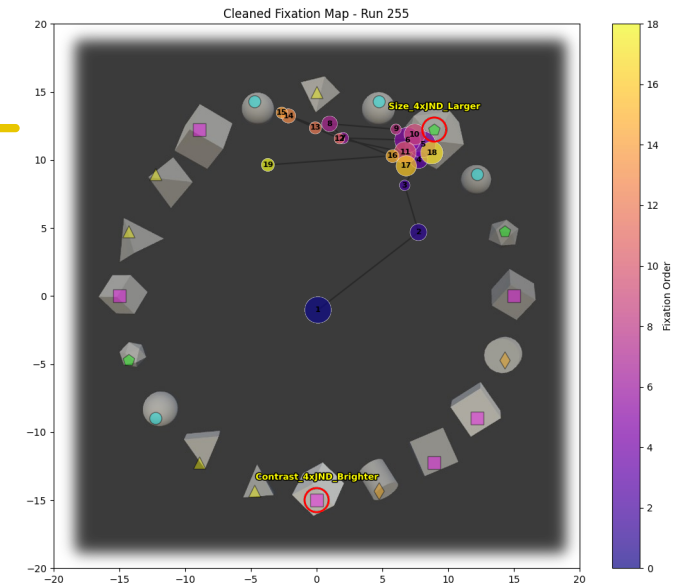
Gaze-Contingent Visual Variables in VR

Idea: Investigate visual variables and gaze behaviour/attention to

- ... gain insights on visual hierarchy
- ... find subtle, yet effectively gaze-capturing variables
- ... improve attention guidance

Setup: Three visual variables: Contrast, Depth, Size

- ... applied to varying objects placed in a circle
- ... in 6 variations based on the JND ($\pm 1x$, $\pm 2x$, $\pm 4x$)
- ... shown separately and in all unique pairwise combinations





Bottom-up and Top-down attention

Eye movements actually **change depending on the question** asked to the subject (task)

Bottom-up attention: involuntary, very quick, task-independent.

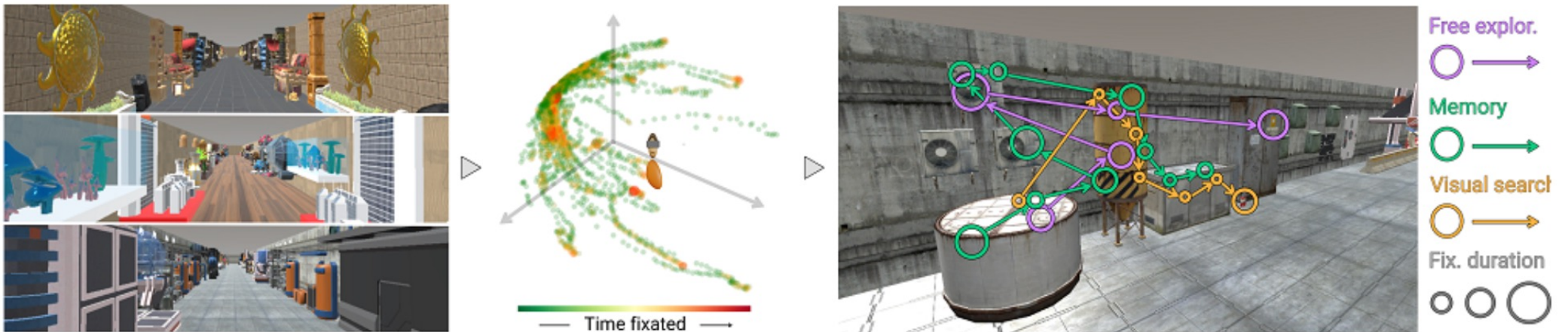
Top-down attention: voluntary, slow, task-dependent.



What we look at depends on what we are trying to do

Top-Down Attention

Task-dependent Visual Behavior in Immersive Environments



Ideas for Open Research Topics

- **Multimodality:** Visual and auditory content (and tactile, olfactory...)
- **Scene dynamics:** Changes in the environment (movement, new stimuli)
- **Task:** Free exploration, memory, search...
- **Scene complexity and cognitive load**
- **Emotional context**
- **User's experience, expectations, familiarity**
- **Diversity!**

...And how all these are entangled!

Ideas for Open Research Topics

- A VR platform to collect training data for attention modelling
- Model scan paths and saccades from low-level visual features (integrating task-description)
- Multi-modal integration of perceptual dimensions (Bottom-up or top-down)
- Machine-learning classifier to distinguish scan-paths and gaze-behaviors
- VLLM for scene analysis and task dependent prediction of attention
- ...

Gaze-Guidance

Idea: Use what we learn from perception-driven rendering and turn it around

Virtual Reality allows us to immerse ourselves in simulated worlds

→ Relevant information are not always in the user's focus

→ Gaze-guidance



Gaze-Guidance

Overt techniques

[RP17, Grü+18, Wen+24]

- + Allows for an immediate recognition
- Can occlude content
- Can negatively impact immersion

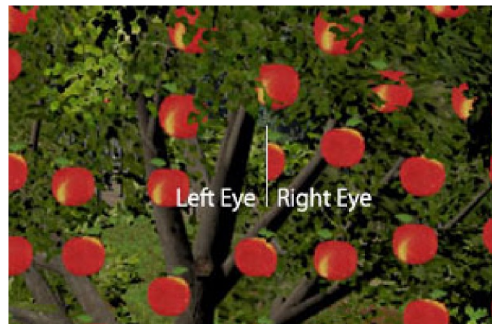


Arrow. Taken from [YKB19].

Subtle techniques

[Gro+20, Kre+20, Tos+25]

- + Low to no negative impact on immersion
- Lower efficiency



DeadEye. Taken from [Lan+20].

Diegetic techniques

[Lan+20, Nor+21]

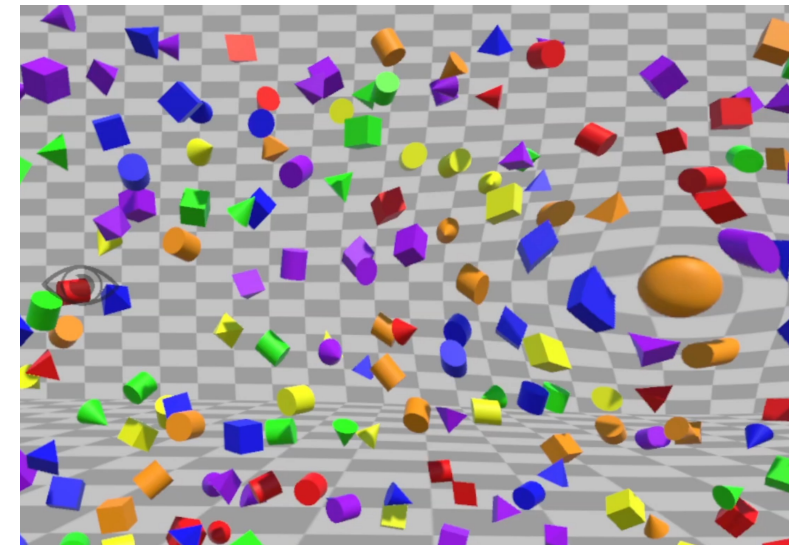
- + Overt, but part of the virtual world
- No general solution for all virtual worlds
- Not always usable for out-of-view guidance



HiveFive. Taken from [Lan+20].

WarpVision

Kudnick, J.; **Weier, M.**; Groth, C.; Fu, B.; Horst, R.; WarpVision: Using Spatial Curvature to Guide Attention in Virtual Reality. IEEE International Symposium on Mixed and Augmented Reality (ISMAR), IEEE TVCG, 2025.



Spatial curvature used to guide gaze

- Distorts the target object and its surrounding
- Distortion is based on the user's gaze-contingent distance to the target object

RippleVision

Waves

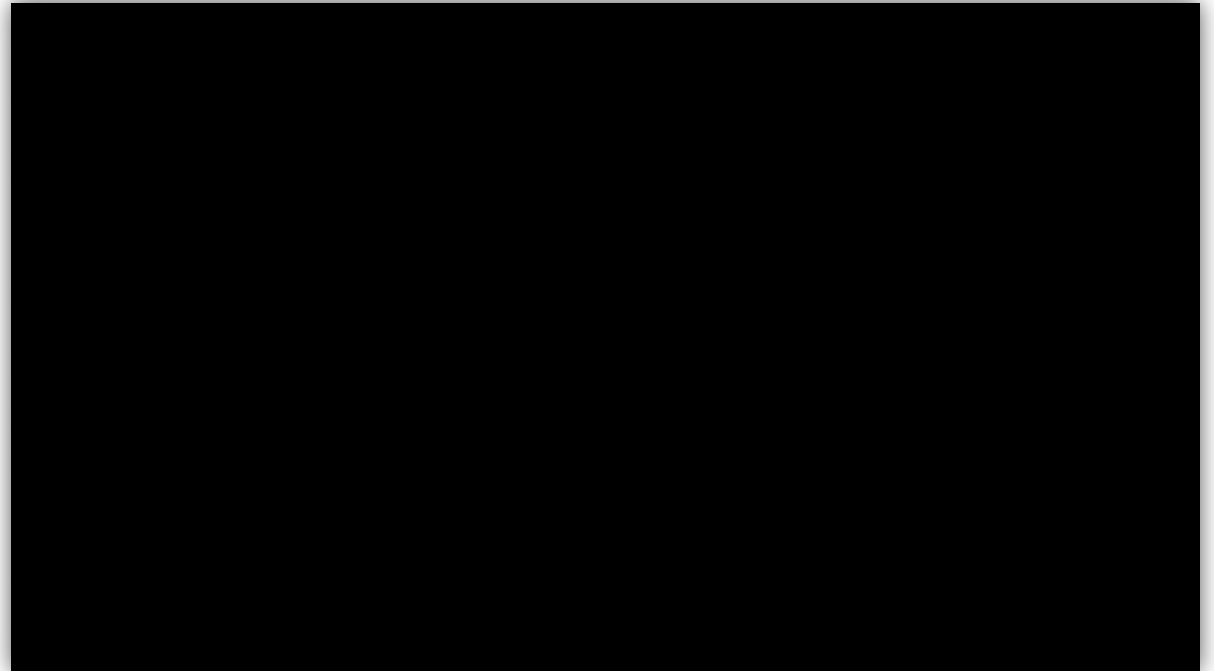
- ... propagate from gaze point (GP) to POI in circular slice
- ... are expressed as modulation in brightness – phase-shifted for each eye

Parameters:

- Intensity (brightness)
 - Wavelength
 - Velocity
 - Angle (size of circular slice)
- Adjusted via calibration study (7m, 1f)

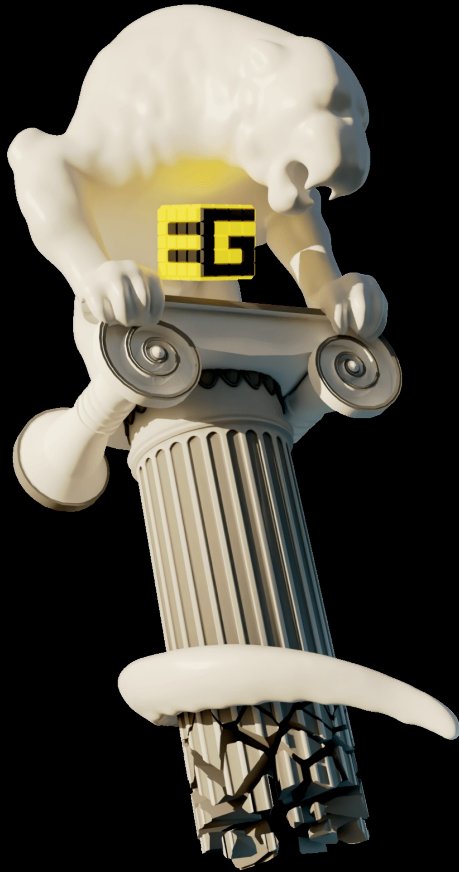
Type:

- Depending on intensity subtle to overt
- Non-diegetic
- Full coverage of out-of-view objects
- Initially indirect



Ideas for Open Research Topics

- Guiding people through walkable VR Worlds
- Guiding metaphors in AR Systems
- Automatic diegetic guidance for Virtual Worlds and XR systems
- Multi-model integration with audio and haptics
- ...



Thank you!

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hsrm 
Design Informatik
Medien



<https://github.com/XRG-HSRM>